USING AI FOR LEARNING

SCIENTIFIC ENGLISH

Especially for students

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Compiled and adapted for educational purposes by:
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Using AI for learning **Scientific [language]**

Tools

Completing a Prompt Engineering course is highly advised! Register with ChatGPT on your computer or use Bing on mobile OR as add-on to Edge browser and engage as follows.

Introduction

GPT-3.5 is an AI language model that, like GPT-4, accesses information and generates content in multiple natural languages. However, the availability and quality of training data can vary across languages, with languages like English often having more extensive datasets. Keep in mind that the prompts in the course were tested in the English language using GPT-3.5. Depending on the language model used and the Input language, you might get different outputs, and indeed, entering the very same prompt might also result in a bit different output - as it is a “generative” system - each time generating an output.

One of the aims of this course is to raise learners' competences in scientific English, so that they could understand better written English text and hopefully also be able to write better prompts. Nevertheless, this depends on the persistence of the learner with the use of prompt combinations offered in this area!. Mastering prompting allows participants to explore topics further and engage in deeper knowledge that is relevant to their own interest or profession as well as improve skills in scientific English.

Progressing through the prompt patterns requires motivation and diligence in reading the prompt constructs and practising alongside, by having another window opened where the actual prompts could be tested, the feedback reviewed and the prompts further adapted to needs. The level of scientific English can be tested through the EnSmartAsses API (in the left menu), providing feedback for progress.

NOTE: The use of LLMs should be considered as a partner in learning (not a teacher), as this technology is still in a very early stage, thus the result of prompts cannot be taken for granted and needs critical thinking and decision making, whether we trust results or not, and needs fact checking. Beware of submitting output from LLMs as your own creation, without careful consideration, as you are the sole responsible person if parts of the date contains untrue, biassed or someone else's copyrighted work.

Learning Scientific [language]

Basic level

Experiment with the following prompts in order to improve your scientific language skills:

Getting advice for improving language skills:

> **Generate a list of specific and actionable steps that I can take to improve my [language].**

Getting advice for improving scientific language skills:

> **How can you help me to improve my scientific [language] in [field].**

Getting a precise scientific language translator:

> **Act as an [[language] to [language]] translator. From now on, if I write text in [language], translate it precisely into scientific [language] text.**

Getting definitions of keywords from a provided text:

> **Give me 5 key words from the following piece of text [paste text] and provide definitions for all of them.**

Getting professional explanation in plain English of an input text:

> **Act as a [field] expert and explain to me in detail the precise meaning of the text I input. Be aware that I am not an expert in [field], also use simple [language] in your explanation.**

Getting professionally recommended resources for learning a topic/skill::

> **Can you suggest expert-recommended resources to learn about [topic/skill]**

Getting plain English explanations for beginners, while investigating understanding and giving further advice:

> **Explain [topic or skill] in the simplest terms possible as if teaching it to a complete beginner. Identify gaps in my understanding and suggest resources to fill them.**

Self-assessment

Getting help for studying the glossary of terms:

Generate a glossary of terms and definitions on [concept].

If you wish the list to continue, just prompt: „Continue” till you get enough terms. After looking through the provided material and reading the meaning of each term, you can test your gained knowledge with some of the following prompts:

- **Generate a self-test for understanding the above and also provide the answers.**
- **Generate fill-in-the blank statements one at a time for practising the use of the terms above. When I give the required term, provide me with detailed feedback.**
- **Generate a self-test for understanding the connections between the above terms and also provide the answers.**

Getting mentoring feedback on understanding of concepts:

Generate simple questions one at a time on [concept]. Whenever I provide an answer, provide me with corrections, also explain with simple terms the reasons the corrections are needed.

Getting continuous mentoring feedback on understanding of concepts and advice for improvement:

Create questions accessing my knowledge of [concept] one-by-one. When I provide an answer, identify areas for improvement.

Getting adaptive mentoring feedback on self-assessment:

Generate a self-assessment task. I'm trying to improve my understanding of [concept]. Ask me a question on [concept], and keep asking me adaptive questions for me to improve.

If you get the answer right, it will give you a more difficult question, if you get it wrong, it will provide you with feedback and give you an easier question.

Getting multiple choice questions on a transcript of a youtube video:

Create questions based on a video transcript: [Copy YouTube (or other video) transcript and paste here]. Generate 5 multiple choice/fill-in-the-blank/open-ended questions from this transcript.

Intermediate level

Experiment with the following prompts in order to further improve your scientific language skills.

Getting a summary of given text:

> **Can you provide a summary of the main points from the following text: [pasted text]**

Getting mentoring behaviour on an expert topic:

> **You are an expert in [topic], and I want you to be my mentor.**

Getting a speedy study schedule studying a specific topic to fit a given job:

> **I want to learn [topic] in [1 months], and my goal would be to [acquire the necessary knowledge to get my first job in the [data] area], can you design a study schedule? What are the things I should learn if I have 1 hour per day, 5 days a week, and I want to learn as quickly as possible.**

Getting a language mentor for intermediate scientific English study:

> **Act as a language teacher for scientific English and provide me with a short scientific text on [concept]. Then ask me questions one-by-one to make sure I understood the text. If I answer correctly, then praise me. If there is space for improvement, then encourage me by explaining how to improve.**

Getting a proof-reader for input text and advice for improvements:

> **I want you to act as a proof-reader. I will provide you with text and I would like you to review them for any spelling, grammar, or punctuation errors. Once you have finished reviewing the text, provide me with any necessary corrections or suggestions for improvement.**

Getting a teacher specialised in topic for testing common phrases:

> **You are my [topic] teacher. Can you test me on [some common phrases] I should know?**

Getting a teacher specialised in topic for asking questions on common phrases with feedback:

> **You are my [topic] teacher. Can you test me on [some common phrases] I should know. Ask me a question, and I'll give you an answer. Then you give me a response. After that, ask me another question. Let's do this 5 times.**

Getting media titles for learning topic:

> **I want to learn how to learn [topic]. I am a visual learner and prefer to [watch videos]. Can you give me [5] resources I can use to help learn.**

Engage in conversation:

> **Engage me in a conversation in the [language] language, helping me to continuously improve.**

Getting a teacher to build skills for learning from scratch topic:

> **I'm learning [topic] and I'm a beginner. What exercises or projects can I work on that will help me build my skills? Give me 3 simple ones.**

Getting a teacher providing analogies for concepts:

> **I want you to come up with 10 simple analogies to illustrate the fact that [concept].**
> **[guidelines] is like...**

Getting sceptical feedback from scientific filed concerning given statements, with explanations:

> **Act as a sceptic scientist in [field]. I will provide you with statements and you will provide sceptical answers with detailed explanations.**

If you wish to learn about specific topic a bit more deeper, then you can construct prompts in the following manner:

Getting a teacher on Machine Learning who explains for beginners and practitioners:

> **I want you to act as a Machine Learning Educator, providing a comprehensive introduction to the core concepts and techniques used in the field. Explain the differences between supervised, unsupervised, and reinforcement learning, and provide examples of real-world applications for each approach. Discuss the importance of feature engineering, model selection, and validation in the development of effective machine learning models. As an educator, present this information in a clear, accessible manner, making it engaging and informative for both beginners and experienced practitioners.**

Getting a specialist on Quantum Computing who briefs on state-of-the art trends for the future:

> **I want you to act as a Quantum Computing specialist, delving into the principles and potential applications of quantum computing. Explain the key differences between classical and quantum computing, focusing on qubits, superposition, and entanglement. Discuss current research and advancements in the field, as well as the potential impact of quantum computing on cryptography, optimization, and artificial intelligence. As a specialist, provide a thorough yet accessible overview of this cutting-edge technology and its implications for the future.**

Getting an expert on Natural Language Processing exploring research directions:

> **I want you to act as a Natural Language Processing expert, describing the core techniques and applications of NLP in the field of AI. Discuss topics such as tokenization, part-of-speech tagging, sentiment analysis, and machine translation. Explain the role of deep learning and transformer models, such as BERT and GPT, in advancing the state of the art in NLP. As an expert, provide insights into current challenges and future directions for natural language processing research and applications.**

Getting a specialist on Cybersecurity Analytics explaining how AI could protect from digital threats:

> **I want you to act as an AI in Cybersecurity Analyst, examining the role of artificial intelligence in detecting and preventing cyber threats. Discuss the use of AI-driven technologies, such as anomaly detection, threat hunting, and automated incident response. Explore the potential risks and challenges associated with using AI in cybersecurity, including the possibility of adversarial attacks and the need for ongoing system updates. Offer suggestions for how AI can be effectively integrated into cybersecurity strategies to protect organisations and individuals from digital threats.**

Getting a professional of Robotic Engineering explaining opportunities and challenges for different industries:

> **I want you to act as a Robotics Engineer, providing an overview of the current state of robotics and its potential future applications. Describe the key components and technologies involved in designing and building robots, including sensors, actuators, and control systems. Discuss the role of AI in enabling more advanced, autonomous robots capable of complex tasks and decision-making. As an engineer, present both the opportunities and challenges associated with the development and deployment of robotic systems in various industries and settings.**

Getting a professional in Computer Vision explaining opportunities and challenges for different industries and sectors:

> **I want you to act as a Computer Vision Specialist, providing a detailed overview of the techniques and applications of computer vision in the field of artificial intelligence. Explain core concepts such as image processing, feature extraction, object recognition, and scene understanding. Discuss the role of deep learning, particularly convolutional neural networks, in advancing the state of the art in computer vision. As a specialist, highlight current challenges and future directions for research and applications, emphasising the potential impact of computer vision technology on various industries and sectors.**

Getting a Sustainability Advocate explaining the possibilities of AI to overcome challenges with cross-disciplinary collaboration:

> **I want you to act as an AI for Sustainability Advocate, exploring how artificial intelligence can be leveraged to address global environmental challenges and promote sustainable practices. Discuss applications of AI in areas such as energy efficiency, climate modelling, conservation efforts, and waste reduction. Highlight the potential benefits and challenges of using AI to tackle these issues, and suggest strategies for ensuring that AI-driven solutions are developed and deployed responsibly. As an advocate, emphasise the importance of cross-disciplinary collaboration and long-term thinking in the pursuit of a more sustainable future.**

English Translator:

> **I want you to act as an English translator, spelling corrector, and improver. I will speak to you in any language and you will detect the language, translate it**

and answer in the corrected and improved version of my text in English. I want you to replace my simplified A0-level words and sentences with more beautiful, elegant, upper-level English words and sentences. Keep the meaning the same, but make them more literary. I want you to only reply to the correction, and the improvements, and nothing else; do not write explanations. My first sentence is "je voudrais un verre de vin"

Ref:@SM Raiyyan https://medium.com/@smraiyyan/awesome-chatgpt-prompts-part-18-185dda4ca5be

Use Bing:

Getting links to learn computers science terminology in English:

Provide me links for learning Scientific English in Computer Science.

Getting the summary of the given book title with outline of concepts:

Summarise the [book] ["Title"] and give me an outline of the most important concepts.

Getting the summary of the given book title with main points:

Summarise this article and give me the main points :
[https://www.nytimes.com/guides/well/how-to-meditate]

Advanced level

Experiment with the following prompts in order to fine-tune your scientific language skills.

Getting insights into both sides of a debate proposed:

> **I want you to act as a debater. I will provide you with some topics related to current events and your task is to research both sides of the debates, present valid arguments for each side, refute opposing points of view, and draw persuasive conclusions based on evidence. Your goal is to help people come away from the discussion with increased knowledge and insight into the topic at hand. My first request is "I want an opinion about ChatGPT in education."**

Getting step-by-step solutions from field expert on given problem:

> **I want you to act as an [field] expert. I will provide you with all the information needed about my technical problems, and your role is to solve my problem. Using intelligent, simple, and understandable language in your answers. It is helpful to explain your solutions step by step and with bullet points. Try to avoid too many technical details but use them when necessary. I want you to reply with the solution, not write any explanations.**

Getting a motivational coach for goals and challenges:

> **I want you to act as a motivational coach. I will provide you with some information about someone's goals and challenges, and it will be your job to come up with strategies that can help this person achieve their goals. This could involve providing positive affirmations, giving helpful advice or suggesting activities they can do to reach their end goal. My first request is "I need help motivating myself to stay disciplined while studying for an upcoming exam".**

Getting career counsellor with background knowledge in job market:

> **I want you to act as a career counsellor. I will provide you with an individual looking for guidance in their professional life, and your task is to help them determine what careers they are most suited for based on their skills, interests, and experience. You should also conduct research into the various options available, explain the job market trends in different industries and advice on which qualifications would be beneficial for pursuing particular fields. My first request is "I want to advise someone who wants to pursue a potential career in software engineering."**

Getting a job interviewer to practise conversation for a job interview:

> **I want you to act as an interviewer. I will be the candidate and you will ask me the interview questions for the [position] position. I want you to only reply as the interviewer. Do not write all the conservation at once. I want you to only do the interview with me. Ask me the questions and wait for my answers. Do not write explanations. Ask me the questions one by one like an interviewer does and wait for my answers. My first sentence is "Hi""**

Getting a plagiarism checker providing uniques versions for given text:

> **I want you to act as a plagiarism checker. I will write you sentences and you will only reply undetected in plagiarism checks in the language of the given sentence, and nothing else. Do not write explanations on replies. My first sentence is "For computers to behave like humans, speech recognition systems must be able to process nonverbal information, such as the emotional state of the speaker."**

Getting advice on modifications of a article to suit review guidelines:

> **I'm going to submit an article that has to meet specific guidelines.**
> **Here are the guidelines:**
> **[guidelines]**
> **Now, here's what I want you to do:**
> **1. Ask me for the article.**
> **2. Then, you'll assess the article based on the guidelines. Give me specific suggestions on how I can improve. Also, give it a score from 0 to 10, where 10 is meeting the guidelines the best for each section. Finally, give it an overall score of how it met the guidelines.**
> **Understood?**

Ref:@DatHero https://medium.com/@eldatero/how-to-use-chatgpt-to-learn-any-skill-quickly-2efd77756663

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